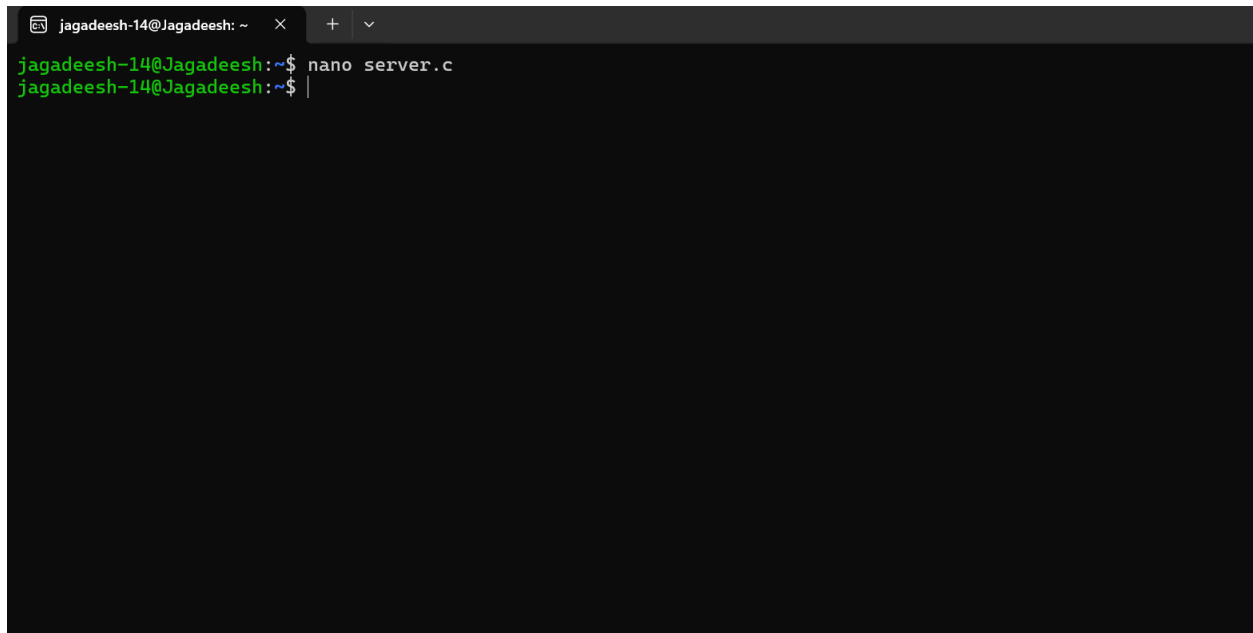




```
jagadeesh-14@Jagadeesh: ~$ nano server.c
jagadeesh-14@Jagadeesh: ~$
```

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <errno.h>

#define SERVER_FIFO "/tmp/server_fifo"
#define MAX_SIZE 512

int main() {
    int server_fd;
    char buffer[MAX_SIZE];
```

```
// Create server FIFO

if (mkfifo(SERVER_FIFO, 0666) == -1 && errno != EEXIST) {

    perror("mkfifo");

    exit(1);

}

printf("Server started...\n");

printf("Waiting for clients...\n");


// Open FIFO for reading

server_fd = open(SERVER_FIFO, O_RDONLY);

if (server_fd == -1) {

    perror("open");

    exit(1);

}

while (1) {

    memset(buffer, 0, sizeof(buffer));

    int bytes = read(server_fd, buffer, sizeof(buffer) - 1);

    if (bytes > 0) {

        buffer[bytes] = '\0';

        // Message format: clientPID:message

        char *colon = strchr(buffer, ':');

        if (colon != NULL) {

            *colon = '\0';

            char *client_pid = buffer;

            char *message = colon + 1;
```

```

printf("\nReceived from Client %s: %s\n",
      client_pid, message);

// Create response FIFO for this client
char response_fifo[100];

snprintf(response_fifo, sizeof(response_fifo),
         "/tmp/client_%s_fifo", client_pid);

int response_fd = open(response_fifo, O_WRONLY);

if (response_fd != -1) {
    char response[MAX_SIZE];

```

output:

```

jagadeesh-14@Jagadeesh: ~
jagadeesh-14@Jagadeesh:~$ nano server.c
jagadeesh-14@Jagadeesh:~$ gcc server.c -o server
server.c: In function 'main':
server.c:55:39: warning: '%s' directive output may be truncated
writing up to 511 bytes into a region of size 88 [-Wformat-trunc
ation=]
55 |         "/tmp/client_%s_fifo", client_p
id);
|
server.c:54:17: note: 'snprintf' output between 18 and 529 bytes
into a destination of size 100
54 |         snprintf(response_fifo, sizeof(response_
fifo),
|
55 |         "/tmp/client_%s_fifo", client_p
id);
|
jagadeesh-14@Jagadeesh:~$ ls
PID.c      Sum.c      linux.c    os.c       prog.c     s
kill4task1.c skill4task3.c skill4task5.c task03.c
PIDs.c     client     myfile.txt process     server     s
kill4task2     skill4task4     task01     task1.c
PIDs.c.save client.c notes.txt.save process.c server.c s
kill4task2.c skill4task4.c task01.c task1.c.save
Sum        FIRST.C   os         prog       skill4task1 s
jagadeesh-14@Jagadeesh:~$ ./server
Server started...
Waiting for clients...

Received from Client 699: Hello Server

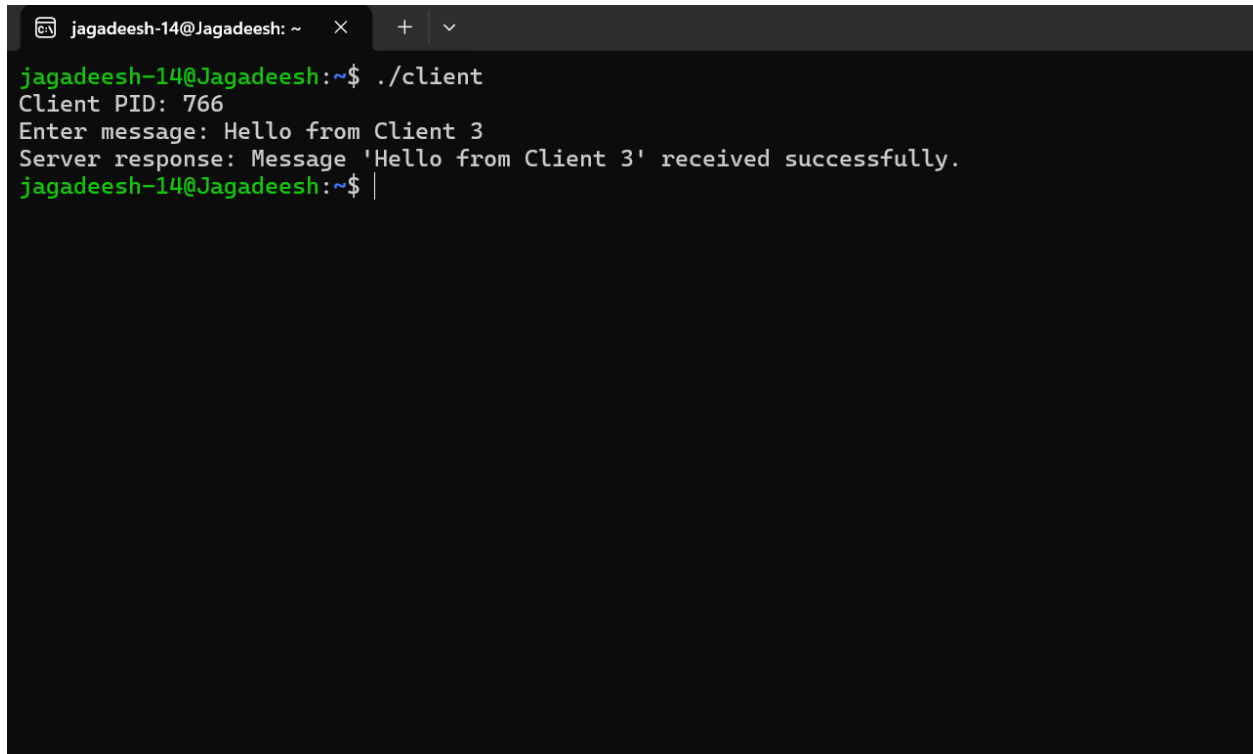
```

```

jagadeesh-14@Jagadeesh:~$ nano client.c
jagadeesh-14@Jagadeesh:~$ gcc client.c -o client
client.c: In function 'main':
client.c:46:18: warning: '%s' directive output may be truncated
writing up to 511 bytes into a region of size between 500 and 51
0 [-Wformat-truncation=]
46 |         "%d:%s", pid, message);
|
client.c:45:5: note: 'snprintf' output between 3 and 524 bytes i
nto a destination of size 512
45 |         snprintf(buffer, sizeof(buffer),
|
46 |         "%d:%s", pid, message);
|
jagadeesh-14@Jagadeesh:~$ ls
PID.c      Sum.c      linux.c    os.c       prog.c     s
kill4task1.c skill4task3.c skill4task5.c task03.c
PIDs.c     client     myfile.txt process     server     s
kill4task2     skill4task4     task01     task1.c
PIDs.c.save client.c notes.txt.save process.c server.c s
kill4task2.c skill4task4.c task01.c task1.c.save
Sum        FIRST.C   os         prog       skill4task1 s
jagadeesh-14@Jagadeesh:~$ ./client
Client PID: 699
Enter message: Hello Server
Server response: Message 'Hello Server' received successfully.
jagadeesh-14@Jagadeesh:~$

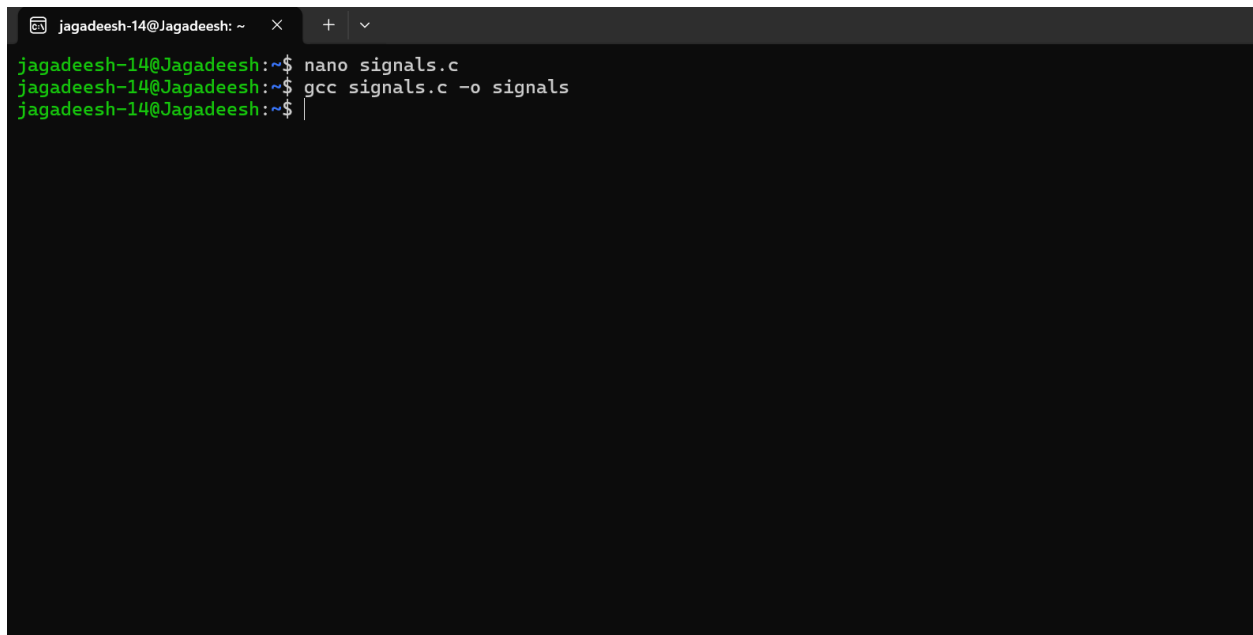
```

Multiple terminals:

A terminal window with a dark background and a title bar that reads 'jagadeesh-14@Jagadeesh: ~'. The terminal shows the following text: a green prompt 'jagadeesh-14@Jagadeesh:~\$' followed by the command './client', the output 'Client PID: 766', the input 'Enter message: Hello from Client 3', the output 'Server response: Message 'Hello from Client 3' received successfully.', and a final green prompt 'jagadeesh-14@Jagadeesh:~\$' with a cursor. The window has standard OS controls (close, maximize, zoom) on the right side of the title bar.

```
jagadeesh-14@Jagadeesh: ~  
jagadeesh-14@Jagadeesh:~$ ./client  
Client PID: 766  
Enter message: Hello from Client 3  
Server response: Message 'Hello from Client 3' received successfully.  
jagadeesh-14@Jagadeesh:~$
```

Task-2:

A terminal window with a dark background and a title bar that reads 'jagadeesh-14@Jagadeesh: ~'. The terminal shows the following text: a green prompt 'jagadeesh-14@Jagadeesh:~\$' followed by the command 'nano signals.c', another green prompt 'jagadeesh-14@Jagadeesh:~\$' followed by the command 'gcc signals.c -o signals', and a final green prompt 'jagadeesh-14@Jagadeesh:~\$' with a cursor. The window has standard OS controls (close, maximize, zoom) on the right side of the title bar.

```
jagadeesh-14@Jagadeesh: ~  
jagadeesh-14@Jagadeesh:~$ nano signals.c  
jagadeesh-14@Jagadeesh:~$ gcc signals.c -o signals  
jagadeesh-14@Jagadeesh:~$
```

```
jagadeesh-14@Jagadeesh: ~  
jagadeesh-14@Jagadeesh:~$ nano signals.c  
jagadeesh-14@Jagadeesh:~$ gcc signals.c -o signals  
jagadeesh-14@Jagadeesh:~$ ./signals  
Signal handling program started.  
Process ID: 808  
Waiting for signals...  
  
jagadeesh-14@Jagadeesh:~$ kill -SIGUSR1 2456  
-bash: kill: (2456) - No such process  
jagadeesh-14@Jagadeesh:~$
```

```
SIGTERM received!  
Termination request received.  
Program terminating...  
Program ended.
```